

SHARP GLOBAL AND ALMOST EVERYWHERE CONVERGENCE RATES FOR PERIODIC HOMOGENIZATION OF VISCOUS QUADRATIC HAMILTON-JACOBI EQUATIONS

ZIRAN LIU, HUNG V. TRAN, YIFENG YU

ABSTRACT. We study the periodic homogenization of the viscous Hamilton–Jacobi equation

$$u_t^\varepsilon + \frac{1}{2}|Du^\varepsilon|^2 + V\left(\frac{x}{\varepsilon}\right) = \frac{\varepsilon}{2}\Delta u^\varepsilon \quad \text{in } \mathbb{R}^n \times (0, \infty),$$

with initial datum $g \in W^{1,\infty}(\mathbb{R}^n)$, where V is Lipschitz continuous and $\mathbb{Z}^n$ -periodic. We prove the sharp global estimate

$$|u^\varepsilon(x, t) - u(x, t)| \leq \varepsilon \left(C + \frac{n}{2} \log \left(\frac{\max\{t, \varepsilon\}}{\varepsilon} \right) \right) \quad \text{for all } (x, t) \in \mathbb{R}^n \times [0, \infty),$$

where $\varepsilon \in (0, 1]$, u solves the limiting (homogenized) equation and $C > 0$ is a constant depending only on $\|Dg\|_{L^\infty(\mathbb{R}^n)}$, $\|DV\|_{L^\infty(\mathbb{R}^n)}$, and n .

We further show that if g is locally semiconcave, then

$$|u^\varepsilon(x, t) - u(x, t)| \leq C_{x,t}\varepsilon \quad \text{for a.e. } (x, t) \in \mathbb{R}^n \times (0, \infty),$$

where $C_{x,t}$ depends on (x, t) , $\|Dg\|_{L^\infty(\mathbb{R}^n)}$, and $\|DV\|_{L^\infty(\mathbb{R}^n)}$. More precisely, the above improved rate holds at every point (x, t) where $u(\cdot, t)$ is twice differentiable at x . In particular, this occurs for a.e. $x \in \mathbb{R}^n$, since $u(\cdot, t)$ is locally semiconcave.

We conclude by raising the open problem of whether the same $O(\varepsilon|\log \varepsilon|)$ rate remains valid for general strictly convex Hamiltonians or general periodic diffusions.

1. INTRODUCTION

For every $\varepsilon > 0$, we consider the Cauchy problem

$$(1) \quad \begin{cases} u_t^\varepsilon + H\left(\frac{x}{\varepsilon}, Du^\varepsilon\right) = \frac{\varepsilon}{2}\Delta u^\varepsilon & \text{in } \mathbb{R}^n \times (0, \infty), \\ u^\varepsilon(x, 0) = g(x) & \text{on } \mathbb{R}^n, \end{cases}$$

and denote by $u^\varepsilon \in C(\mathbb{R}^n \times [0, \infty))$ its viscosity solution. The Hamiltonian $H = H(y, p) : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ and the initial data $g : \mathbb{R}^n \rightarrow \mathbb{R}$ are prescribed. Assume that

- (A1) $H \in \text{Lip}_{\text{loc}}(\mathbb{R}^n \times \mathbb{R}^n)$, and for each $p \in \mathbb{R}^n$, the map $y \mapsto H(y, p)$ is $\mathbb{Z}^n$ -periodic;
(A2)

$$\lim_{|p| \rightarrow \infty} \text{ess inf}_{y \in \mathbb{R}^n} (|H(y, p)|^2 - (n+1)D_y H(y, p) \cdot p) = +\infty;$$

- (A3) $g \in W^{1,\infty}(\mathbb{R}^n)$.

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A basic example of H is the quadratic case (mechanical Hamiltonian):

$$H(y, p) = \frac{1}{2}|p|^2 + V(y) \quad \text{for } (y, p) \in \mathbb{R}^n \times \mathbb{R}^n.$$

Under assumptions (A1)–(A3), it is known that, as $\varepsilon \rightarrow 0+$, the family $\{u^\varepsilon\}$ converges locally uniformly in $\mathbb{R}^n \times [0, \infty)$ to a function $u \in C(\mathbb{R}^n \times [0, \infty))$. Moreover, u is the viscosity solution of the homogenized equation

$$(2) \quad \begin{cases} u_t + \overline{H}(Du) = 0 & \text{in } \mathbb{R}^n \times (0, \infty), \\ u(x, 0) = g(x) & \text{on } \mathbb{R}^n. \end{cases}$$

See [8]. Here $\overline{H} \in C(\mathbb{R}^n)$ denotes the effective Hamiltonian associated with H , defined through the corresponding cell problem: for any $p \in \mathbb{R}^n$, there exists a unique number $\overline{H}(p) \in \mathbb{R}$ such that the following cell problem has a $\mathbb{Z}^n$ -periodic viscosity solution

$$-\frac{1}{2}\Delta v + H(y, p + Dv) = \overline{H}(p) \quad \text{in } \mathbb{R}^n.$$

If H is convex in p , then $\overline{H}$ is also convex in p . In this case, we denote by $\overline{L}$ the effective Lagrangian associated with $\overline{H}$:

$$(3) \quad \overline{L}(q) = \sup_{p \in \mathbb{R}^n} \{q \cdot p - \overline{H}(p)\}.$$

A natural question is to identify the convergence rate of $u^\varepsilon \rightarrow u$ as $\varepsilon \rightarrow 0$, which belongs to the very active area of quantitative homogenizations. For general H satisfying (A1)–(A3), it was proved in [24] that

$$|u^\varepsilon - u| \leq O(\sqrt{\varepsilon}).$$

See [4, 3] for the earlier $O(\varepsilon^{1/3})$ -rate. The argument is based on the perturbed test function method in [8], together with a careful choice of scaling in ε . The above rate turns out to be optimal in the sense that there exist particular choices of H and g satisfying assumptions (A1)–(A3), where H is not strictly convex in p , such that the convergence rate is $\sqrt{\varepsilon}$.

In this paper, we consider the most interesting example $H(y, p) = \frac{1}{2}|p|^2 + V(y)$. The equation (1) becomes

$$(4) \quad \begin{cases} u_t^\varepsilon + \frac{1}{2}|Du^\varepsilon|^2 + V\left(\frac{x}{\varepsilon}\right) = \frac{\varepsilon}{2}\Delta u^\varepsilon & \text{in } \mathbb{R}^n \times (0, \infty), \\ u^\varepsilon(x, 0) = g(x) & \text{on } \mathbb{R}^n. \end{cases}$$

Numerical results in [24] suggest that the convergence rate is faster, approaching ε . In this paper, we will confirm this observation. A new approach is needed since the method used in [24] is too coarse to obtain the optimal convergence rate for the quadratic case.

Below are our main results.

Theorem 1.1. *Assume that $H(y, p) = \frac{1}{2}|p|^2 + V(y)$ for $(y, p) \in \mathbb{R}^n \times \mathbb{R}^n$, where $V \in \text{Lip}(\mathbb{R}^n)$ is $\mathbb{Z}^n$ -periodic. Assume further that $g \in W^{1,\infty}(\mathbb{R}^n)$. Then, there exists*

$C = C(\|Dg\|_{L^\infty(\mathbb{R}^n)}, \|DV\|_{L^\infty(\mathbb{R}^n)}, n) > 0$ such that, for $(x, t) \in \mathbb{R}^n \times [0, \infty)$ and $\varepsilon \in (0, 1)$,

$$(5) \quad |u^\varepsilon(x, t) - u(x, t)| \leq \varepsilon \left(C + \frac{n}{2} \log \left(\frac{\max\{t, \varepsilon\}}{\varepsilon} \right) \right).$$

Theorem 1.2. *Assume that $H(y, p) = \frac{1}{2}|p|^2 + V(y)$ for $(y, p) \in \mathbb{R}^n \times \mathbb{R}^n$, where $V \in \text{Lip}(\mathbb{R}^n)$ is $\mathbb{Z}^n$ -periodic. Assume further that $g \in W^{1,\infty}(\mathbb{R}^n)$ is locally semiconcave. Then, for each $t > 0$, $u(\cdot, t)$ is locally semiconcave and is twice differentiable almost everywhere in $x \in \mathbb{R}^n$.*

For $(x, t) \in \mathbb{R}^n \times (0, \infty)$, if u is twice differentiable at x , then

$$(6) \quad |u^\varepsilon(x, t) - u(x, t)| \leq C_{x,t} \varepsilon$$

for a constant $C_{x,t}$ depending on the point (x, t) , $\|Dg\|_{L^\infty(\mathbb{R}^n)}$, and $\|DV\|_{L^\infty(\mathbb{R}^n)}$.

It is worth to note that Lemma 3.5 in the proof of Theorem 1.2 was supplied by ChatGPT, and verified carefully by the authors.

Remark 1.3. When the diffusion term is absent, the optimal convergence rate is $O(\varepsilon)$ for all convex coercive Hamiltonians H ; see [27]. See [11, 12, 20, 21, 13] and the references therein for optimal rates of first-order problems in various settings.

In the viscous case, the $\varepsilon|\log \varepsilon|$ rate is sharp. See [24, Proposition 4.4] for the matching lower-bound direction in the special case where V is constant (the vanishing viscosity case).

It remains an interesting open problem whether the same $\varepsilon|\log \varepsilon|$ bound holds for other strictly convex Hamiltonians, for example

$$H(p) = \sqrt{1 + |p|^2},$$

which is related to the G -equation with curvature effects in a shear ambient flow; see [30] and also [19]. Likewise, one may ask what the optimal convergence rate is when the term $\varepsilon \Delta u^\varepsilon$ is replaced by a more general diffusion operator of the form $\varepsilon \text{tr} \left(A(\frac{x}{\varepsilon}) D^2 u^\varepsilon \right)$, including the degenerate setting.

We refer the reader to [6, 5, 29] and the references therein for interesting results on the $O(\varepsilon|\log \varepsilon|)$ convergence of the vanishing viscosity process for general strictly convex Hamiltonians.

Remark 1.4. Let us see Theorem 1.1 through a probabilistic (operator) semigroups point of view.

1. For the mechanical Hamiltonian

$$H(y, p) = \frac{1}{2}|p|^2 + V(y),$$

the standard Hopf–Cole transform converts the viscous Hamilton–Jacobi equation into the linear parabolic equation

$$w_t - \frac{1}{2} \Delta w - V(x)w = 0.$$

Equivalently, if one sets

$$\mathcal{L} := \frac{1}{2}\Delta + V,$$

then the transformed solution is governed by the (operator) semigroup $e^{t\mathcal{L}}$:

$$w(\cdot, t) = e^{t\mathcal{L}}w(\cdot, 0).$$

2. For each $p \in \mathbb{R}^n$, the effective Hamiltonian $\overline{H}(p)$ is not only the quantity in the cell problem of periodic homogenization, but also the principal eigenvalue of the tilted periodic operator

$$\mathcal{L}_p \phi = \frac{1}{2}\Delta \phi - p \cdot D\phi + \left(\frac{1}{2}|p|^2 + V(y) \right) \phi \quad \text{in } \mathbb{T}^n.$$

Let $r_p(x) = e^{-v_p(x)}$. Then

$$\mathcal{L}_p r_p = \overline{H}(p) r_p.$$

Equivalently, if $h_p(x) = e^{-p \cdot x - v_p(x)}$, then

$$\mathcal{L} h_p = \overline{H}(p) h_p.$$

The corresponding ground-state, or Doob h -transform, produces the conservative diffusion with generator

$$\mathcal{G}_p f = \frac{1}{2}\Delta f + b_p(x) \cdot Df, \quad b_p(x) = -p - Dv_p(x),$$

whose transition density with respect to Lebesgue measure is denoted by $\tilde{p}_p(t, x, y)$. Accordingly,

$$K(t, x, y) = e^{t\overline{H}(p)} \frac{h_p(x)}{h_p(y)} \tilde{p}_p(t, x, y).$$

3. By the Feynman–Kac formula,

$$(e^{t\mathcal{L}} f)(x) = \mathbb{E}_x \left[\exp \left(\int_0^t V(X_s) ds \right) f(X_t) \right],$$

so that the kernel $K(t, x, y)$ may be viewed as the kernel of a periodic Feynman–Kac weighted diffusion.

Following this probabilistic point of view, one can see that the proof naturally separates into two parts: the exponential scale is encoded by the tilted principal eigenvalue $\overline{H}(p)$, while the sharp pre-exponential information is governed by the long-time ballistic asymptotics of the transformed diffusion kernel $\tilde{p}_p(t, x, y)$.

In such a sense, the argument connects three complementary viewpoints on the same problem: quantitative homogenization for viscous Hamilton–Jacobi equations, semigroup or Feynman–Kac methods and Doob h -transforms, and the spectral description of long-time Gaussian asymptotics in periodic media; see, for example, [7, 15, 17, 18, 22, 23, 25, 14].

Organization of the paper. In Section 2, we review basic properties of the effective Hamiltonian and effective Lagrangian. The proofs of Theorems 1.1 and 1.2 are given in Section 3. A key ingredient is the so-called ballistic estimate" (Proposition 3.1), which is proved in Section 4 as a corollary of [22, Theorem 1.3] via the Doob h -transform. Finally, in Appendix A, for the reader's convenience, we provide a self-contained proof of the large-time Gaussian asymptotics for the heat kernel and derive a sharp ballistic asymptotic formula for the original Schrödinger kernel. Although these more delicate estimates are not needed for our main purpose, both the conclusions and the arguments may be useful in other contexts.

Acknowledgment. Lemma 3.5 in the proof of Theorem 1.2 was supplied by Chat-GPT, and verified carefully by the authors.

2. PRELIMINARIES

In this section, we review some basic properties of the effective Hamiltonian $\overline{H}$. For $p \in \mathbb{R}^n$, let v_p be the unique $\mathbb{Z}^n$ -periodic solution to

$$(7) \quad -\frac{1}{2}\Delta v_p + \frac{1}{2}|p + Dv_p|^2 + V(x) = \overline{H}(p) \quad \text{in } \mathbb{R}^n$$

subject to $v_p(0) = 0$.

Let $\alpha \in (0, 1)$, $\mathbb{T}^n = \mathbb{R}^n / \mathbb{Z}^n$, $X = \{v \in C^{2,\alpha}(\mathbb{T}^n) : v(0) = 0\} \times \mathbb{R}$, and $Y = C^{0,\alpha}(\mathbb{T}^n)$. Apply the Banach implicit function theorem to the map

$$\begin{aligned} \Theta : \mathbb{R}^n \times X &\rightarrow Y \\ \Theta(p, (v, \lambda)) &= -\frac{1}{2}\Delta v + \frac{1}{2}|p + Dv|^2 + V - \lambda, \end{aligned}$$

we have that

$$(8) \quad p \mapsto (v_p, \overline{H}(p)) \quad \text{is } C^\infty.$$

For the reader's convenience, we prove the following properties of $\overline{H}(p)$ that are well known to experts.

Lemma 2.1. *For all $p \in \mathbb{R}^n$,*

- (1) $\overline{H}$ is strictly convex in p , i.e., $D^2\overline{H}(p) > 0$ for $p \in \mathbb{R}^n$;
- (2) $\frac{1}{2}|p|^2 + \min_{\mathbb{R}^n} V \leq \overline{H}(p) \leq \frac{1}{2}|p|^2 + \max_{\mathbb{R}^n} V$ for all $p \in \mathbb{R}^n$;
- (3) $\overline{H}$ is an even function, i.e., $\overline{H}(p) = \overline{H}(-p)$ for $p \in \mathbb{R}^n$.

Proof. We prove (1) by establishing a more general statement. Consider the cell problem

$$-\frac{1}{2}\Delta v_p + H(p + Dv_p) + V(x) = \overline{H}(p) \quad \text{in } \mathbb{R}^n$$

for a smooth and strictly convex function $H = H(p)$. In our case, $H(p) = \frac{1}{2}|p|^2$.

In view of (8), differentiating the above equation twice with respect to p in the direction $\xi \in \mathbb{R}^n$, we obtain

$$-\frac{1}{2}\Delta w + q \cdot D^2 H(p + Dv_p)q + DH(p + Dv_p) \cdot Dw = \xi \cdot D^2 \overline{H}(p)\xi.$$

Here, $w(x) = \xi \cdot D_p^2 v_p(x) \xi$ and $q(x) = (I_n + D_p^2 v_p(x)) \xi$.

Since $q \cdot D^2 H(p + Dv_p) q \geq 0$, if $\xi \cdot D^2 \bar{H}(p) \xi \leq 0$, the strong maximum principle and $w(0) = 0$ imply that

$$w \equiv 0 \quad \text{and} \quad q \cdot D^2 H(p + Dv_p) q = \xi \cdot D^2 \bar{H}(p) \xi = 0.$$

So $q = 0$. Since $\xi \cdot q = |\xi|^2 + w$, we deduce that $\xi = 0$. Hence $D^2 \bar{H}(p) > 0$.

Claim (2) follows immediately by considering the maximum and minimum points of v_p .

We now prove (3). For $p \in \mathbb{R}^n$, let $r_p = e^{-v_p}$. Then

$$\begin{cases} (-\frac{1}{2}\Delta + p \cdot D - V)r_p = E_p r_p & \text{in } \mathbb{T}^n, \\ (-\frac{1}{2}\Delta - p \cdot D - V)r_{-p} = E_{-p} r_{-p} & \text{in } \mathbb{T}^n. \end{cases}$$

Here $E_p = \frac{1}{2}|p|^2 - \bar{H}(p)$. Note that the above two operators are dual. Then $E_p = E_{-p}$. This leads to $\bar{H}(p) = \bar{H}(-p)$. □

As an immediate corollary, we have that

Corollary 2.2. *Let $\bar{L}$ be the corresponding effective Lagrangian. Then:*

- (1) $\bar{L} \in C^\infty(\mathbb{R}^n)$ is strictly convex ($D^2 \bar{L} > 0$) and even in q ;
- (2) for every $q \in \mathbb{R}^n$,

$$\frac{1}{2}|q|^2 - \max_{\mathbb{R}^n} V \leq \bar{L}(q) \leq \frac{1}{2}|q|^2 - \min_{\mathbb{R}^n} V.$$

Now let

$$(9) \quad \pi_p = C r_p r_{-p}.$$

The constant $C > 0$ is chosen so that $\int_{\mathbb{T}^n} \pi_p dx = 1$. This is the same as the measure σ defined in [9].

Lemma 2.3. *We have that*

$$-\frac{1}{2}\Delta \pi_p - \operatorname{div}((p + Dv_p)\pi_p) = 0 \quad \text{in } \mathbb{T}^n.$$

Proof. Since C is a normalization constant, it suffices to prove the result for

$$\pi := r_p r_{-p}.$$

Recall that

$$\left(-\frac{1}{2}\Delta + p \cdot D - V\right) r_p = E_p r_p, \quad \left(-\frac{1}{2}\Delta - p \cdot D - V\right) r_{-p} = E_p r_{-p},$$

with $E_{-p} = E_p$, and

$$r_p = e^{-v_p}, \quad Dr_p = -r_p Dv_p.$$

Hence

$$p + Dv_p = p - \frac{Dr_p}{r_p},$$

and therefore

$$(p + Dv_p)\pi = \left(p - \frac{Dr_p}{r_p}\right)r_p r_{-p} = pr_p r_{-p} - r_{-p} Dr_p.$$

Next,

$$\frac{1}{2}\Delta(r_p r_{-p}) = \frac{1}{2}r_{-p}\Delta r_p + \frac{1}{2}r_p\Delta r_{-p} + Dr_p \cdot Dr_{-p},$$

while

$$\operatorname{div}(pr_p r_{-p} - r_{-p} Dr_p) = r_{-p}p \cdot Dr_p + r_p p \cdot Dr_{-p} - Dr_{-p} \cdot Dr_p - r_{-p}\Delta r_p.$$

Subtracting the two identities gives

$$\frac{1}{2}\Delta\pi + \operatorname{div}((p + Dv_p)\pi) = -r_{-p} \left(\frac{1}{2}\Delta r_p - p \cdot Dr_p\right) + r_p \left(\frac{1}{2}\Delta r_{-p} + p \cdot Dr_{-p}\right).$$

By the equations for r_p and r_{-p} ,

$$\frac{1}{2}\Delta r_p - p \cdot Dr_p = -(V + E_p)r_p, \quad \frac{1}{2}\Delta r_{-p} + p \cdot Dr_{-p} = -(V + E_p)r_{-p}.$$

Substituting these two identities, we obtain

$$\frac{1}{2}\Delta\pi + \operatorname{div}((p + Dv_p)\pi) = 0.$$

Multiplying by C , we conclude that

$$\frac{1}{2}\Delta\pi_p + \operatorname{div}((p + Dv_p)\pi_p) = 0 \quad \text{in } \mathbb{T}^n.$$

□

Lemma 2.4. *For every $p \in \mathbb{R}^n$,*

$$D\overline{H}(p) = \int_{\mathbb{T}^n} (p + Dv_p)\pi_p dx.$$

Proof. For $j \in \{1, \dots, n\}$, let $w_j := \partial_{p_j} v_p$. Differentiating the cell problem (7) with respect to p_j gives

$$-\frac{1}{2}\Delta w_j + (p + Dv_p) \cdot (e_j + Dw_j) = \partial_{p_j} \overline{H}(p) \quad \text{in } \mathbb{T}^n.$$

Multiplying by π_p , integrating over $\mathbb{T}^n$, and using Lemma 2.3, we obtain

$$\partial_{p_j} \overline{H}(p) = \int_{\mathbb{T}^n} (p_j + \partial_j v_p)\pi_p dx.$$

Since this holds for every j , the conclusion follows. □

We refer the reader to [10, 9] for further interesting identities involving π_p and v_p .

3. PROOF OF THEOREMS 1.1 AND 1.2

Owing to translation and scaling, we may reduce the proof of (5) to the case $(x, t) = (0, 1)$. We first apply the Hopf–Cole transform

$$u^\varepsilon(x, t) = -\varepsilon \log w \left(\frac{x}{\varepsilon}, \frac{t}{\varepsilon} \right).$$

This converts the equation (4) to a linear equation satisfied by w :

$$\begin{cases} w_t - \frac{1}{2}\Delta w - V(x)w = 0 \\ w(x, 0) = e^{-\frac{g(\varepsilon x)}{\varepsilon}}. \end{cases}$$

Therefore

$$(10) \quad u^\varepsilon(0, 1) = -\varepsilon \log \left(\varepsilon^{-n} \int_{\mathbb{R}^n} K \left(\frac{1}{\varepsilon}, 0, \frac{y}{\varepsilon} \right) e^{-\frac{g(y)}{\varepsilon}} dy \right).$$

Here $K(t, x, y)$ is the fundamental solution of the Schrödinger operator $\partial_t - \mathcal{L}$ for

$$\mathcal{L} = \frac{1}{2}\Delta + V.$$

The next step is to provide an accurate estimate of the asymptotic behavior of $K(t, x, y)$. By the Feynman–Kac formula,

$$K(t, x, y) = p_t(x - y) \mathbb{E}_t^{x \rightarrow y} \left[\exp \left(\int_0^t V(B_s) ds \right) \right],$$

where the expectation is over the Brownian bridge from x to y in time t . This leads to the following rough estimate of K .

$$(11) \quad e^{-t\|V\|_{L^\infty}} p_t(x - y) \leq K(t, x, y) \leq e^{t\|V\|_{L^\infty}} p_t(x - y),$$

where

$$p_t(z) = \frac{1}{(2\pi t)^{n/2}} e^{-\frac{|z|^2}{2t}}.$$

To obtain the $\varepsilon |\log \varepsilon|$ convergence rate, we need the following *ballistic estimate*.

Proposition 3.1 (Ballistic kernel bounds on compact velocity sets). *Let $V_0 \subset \mathbb{R}^n$ be compact. Then there exist constants $t_0 \geq 1$ and $0 < C_1 \leq C_2 < \infty$ such that, for every $q \in V_0$ and every $t \geq t_0$,*

$$(12) \quad \frac{C_1}{t^{n/2}} e^{-t\bar{L}(q)} \leq K(t, 0, -qt) \leq \frac{C_2}{t^{n/2}} e^{-t\bar{L}(q)}.$$

Remark 3.2. For the proof of Theorem 1.1, only the two-sided bounds (12) are needed. Sharper asymptotic formulas for $K(t, x, y)$ were already available earlier: the one-dimensional case was established in [28, Theorem 1.1], and higher-dimensional large-deviation asymptotics were announced in [1] and later proved in the more general branching-diffusion setting in [14, Theorem 2.2]. Proposition 3.1 should therefore be viewed only as the weaker corollary tailored to the present argument. In Appendix A, we record a sharper asymptotic, Proposition A.3, obtained by our Bloch–Floquet/Doob-transform approach.

Remark 3.3. The Hopf–Cole transform linearizes (4) into a parabolic/Schrödinger equation with potential V as a technique in PDE. Probabilistically, the kernel $K(t, x, y)$ turns out to be the kernel of a Feynman–Kac weighted diffusion, as we have explained in Remark 1.4. Thus the ballistic estimate above may be understood as a precise long-time asymptotic for a weighted diffusion kernel. After the Doob h -transform given in Subsection 4.1, this weighted kernel becomes the heat kernel of a genuine periodic diffusion.

We postpone the proof of Proposition 3.1 to Section 4.

3.1. Proof of Theorem 1.1. Before presenting the proof of Theorem 1.1, we need one final preparatory step.

Lemma 3.4. *There exists $C = C(\|DV\|_{L^\infty}, \|Dg\|_{L^\infty}, n) > 0$ such that, for $(x, t) \in \mathbb{R}^n \times [0, \infty)$,*

$$(13) \quad |u(x, t) - g(x)| \leq Ct \quad \text{and} \quad |u^\varepsilon(x, t) - g(x)| \leq C(t + \varepsilon).$$

Proof. We only need to prove the second inequality in (13), which follows [24, Section 3.2]. Let $\rho \in C_c^\infty(\mathbb{R}^n, [0, \infty))$ be a standard mollifier, that is,

$$\int_{\mathbb{R}^n} \rho(x) dx = 1, \quad \text{supp}(\rho) \subset B(0, 1), \quad \rho(x) = \rho(-x) \text{ for } x \in \mathbb{R}^n,$$

We set $\rho^\varepsilon := \frac{1}{\varepsilon^n} \rho(\frac{\cdot}{\varepsilon})$ and $g^\varepsilon := \rho^\varepsilon * g$. Then, $g^\varepsilon \in C^2(\mathbb{R}^n)$ and we have the bounds

$$(14) \quad \|g^\varepsilon - g\|_{L^\infty(\mathbb{R}^n)} \leq C\varepsilon, \quad \|Dg^\varepsilon\|_{L^\infty(\mathbb{R}^n)} + \varepsilon\|D^2g^\varepsilon\|_{L^\infty(\mathbb{R}^n)} \leq C.$$

Let $\tilde{u}^\varepsilon$ denote the viscosity solution to

$$(15) \quad \begin{cases} \tilde{u}_t^\varepsilon + H\left(\frac{x}{\varepsilon}, D\tilde{u}^\varepsilon\right) = \varepsilon\Delta\tilde{u}^\varepsilon & \text{in } \mathbb{R}^n \times (0, \infty), \\ \tilde{u}^\varepsilon(x, 0) = g^\varepsilon(x) & \text{on } \mathbb{R}^n. \end{cases}$$

By (14) and the comparison principle, we deduce

$$(16) \quad \|\tilde{u}^\varepsilon - u^\varepsilon\|_{L^\infty(\mathbb{R}^n \times [0, \infty))} \leq C\varepsilon.$$

On the other hand, in view of (14), we have

$$\left| H\left(\frac{x}{\varepsilon}, Dg^\varepsilon(x)\right) - \varepsilon\Delta g^\varepsilon(x) \right| \leq C \quad \text{for } x \in \mathbb{R}^n,$$

which yields that $(x, t) \mapsto g^\varepsilon(x) + Ct$ is a supersolution to (15), and $(x, t) \mapsto g^\varepsilon(x) - Ct$ is a subsolution to (15). By the comparison principle,

$$g^\varepsilon(x) - Ct \leq \tilde{u}^\varepsilon(x, t) \leq g^\varepsilon(x) + Ct \quad \text{for all } (x, t) \in \mathbb{R}^n \times [0, \infty),$$

Thus, for $(x, t) \in \mathbb{R}^n \times [0, \infty)$,

$$|u^\varepsilon(x, t) - g(x)| \leq |\tilde{u}^\varepsilon(x, t) - g(x)| + \|\tilde{u}^\varepsilon - u^\varepsilon\|_{L^\infty(\mathbb{R}^n \times [0, \infty))} \leq C(t + \varepsilon).$$

□

We are ready to prove our main results.

Proof of Theorem 1.1. Since replacing V by $V + c$ changes both u^ε and u by the same additive term $-ct$, we may normalize

$$\min_{\mathbb{T}^n} V = 0.$$

Then $\|V\|_{L^\infty(\mathbb{T}^n)} \leq \sqrt{n}\|DV\|_{L^\infty(\mathbb{T}^n)}$, so every constant below may be taken to depend only on n , $\|DV\|_{L^\infty(\mathbb{T}^n)}$, $\|Dg\|_{L^\infty(\mathbb{R}^n)}$.

Fix $(x_0, t_0) \in \mathbb{R}^n \times [0, \infty)$. If $t_0 \leq \varepsilon$, then by Lemma 3.4, we have

$$|u^\varepsilon(x_0, t_0) - u(x_0, t_0)| \leq C(t_0 + \varepsilon) \leq C\varepsilon,$$

which gives us the desired conclusion.

We now consider the case $t_0 \geq \varepsilon$. Let $\tilde{\varepsilon} = \varepsilon/t_0 \in (0, 1]$ and

$$w^{\tilde{\varepsilon}}(x, t) := \frac{1}{t_0} u^\varepsilon(x_0 + xt_0, tt_0), \quad \tilde{g}(x) := \frac{1}{t_0} g(x_0 + xt_0), \quad \tilde{V}(y) := V\left(y + \frac{x_0}{\varepsilon}\right).$$

Then $w^{\tilde{\varepsilon}}$ solves the same viscous Hamilton–Jacobi equation

$$\begin{cases} w_t^{\tilde{\varepsilon}} + \frac{1}{2}|Dw^{\tilde{\varepsilon}}|^2 + \tilde{V}\left(\frac{x}{\tilde{\varepsilon}}\right) = \frac{\tilde{\varepsilon}}{2}\Delta w^{\tilde{\varepsilon}} & \text{in } \mathbb{R}^n \times (0, \infty), \\ w^{\tilde{\varepsilon}}(x, 0) = \tilde{g}(x) & \text{on } \mathbb{R}^n. \end{cases}$$

And the corresponding homogenized solution is $w(x, t) = \frac{1}{t_0} u(x_0 + xt_0, tt_0)$. Since $\tilde{V}$ is again $\mathbb{Z}^n$ -periodic, $\|D\tilde{V}\|_{L^\infty} = \|DV\|_{L^\infty}$, $\|D\tilde{g}\|_{L^\infty} = \|Dg\|_{L^\infty}$, and the effective Hamiltonian is unchanged by spatial translation of the potential, the argument below applies verbatim with the same constants. Hence it suffices to establish the estimate at $(x_0, t_0) = (0, 1)$.

Denote by

$$h(y) := g(y) + \bar{L}(-y) \quad \text{for } y \in \mathbb{R}^n.$$

By the Hopf–Lax formula (see [26, Chapter 2]),

$$u(0, 1) = \min_{\mathbb{R}^n} h = h(\bar{y}),$$

where $\bar{y} \in \mathbb{R}^n$ is a global minimum point of h . Since $g \in W^{1,\infty}(\mathbb{R}^n)$ and $\bar{L}$ grows quadratically thanks to Corollary 2.2, there exists $C = C(\|Dg\|_{L^\infty}, \|DV\|_{L^\infty}, n)$ such that

$$|\bar{y}| \leq C.$$

Indeed, $h(0) \leq g(0)$, while the lower quadratic bound in Corollary 2.2 and the Lipschitz continuity of g imply $h(y) \rightarrow +\infty$ as $|y| \rightarrow \infty$. Furthermore, there exists $A = A(\|Dg\|_{L^\infty}, \|DV\|_{L^\infty}) > 0$ such that

$$(17) \quad |h(y) - h(\bar{y})| \leq A|y - \bar{y}| \quad \text{for } y \in B(\bar{y}, 20C).$$

Indeed, since $\bar{L} \in C^1(\mathbb{R}^n)$ and $q \mapsto D\bar{L}(q)$ is bounded on the compact set $\bar{B}(0, 20C)$, the Lipschitz constant of h on $B(\bar{y}, 20C)$ is bounded only in terms of $\|Dg\|_{L^\infty}$, $\|DV\|_{L^\infty}$, and n .

Recall that the Hopf–Cole representation (10) and the rough kernel bound (11) have already been established just before the proof. Let $V_0 = \bar{B}(0, 20C)$. Since $K(s, 0, -qs)$

is strictly positive and continuous in (s, q) for $s > 0$, and since $\bar{L}$ is continuous, the quantity

$$s^{n/2} e^{s\bar{L}(q)} K(s, 0, -qs)$$

is bounded above and below by positive constants on the compact set $[1, t_0] \times V_0$. Combining this with Proposition 3.1, and enlarging the constants if necessary, we obtain positive constants c_τ and C_τ such that

$$(18) \quad \frac{c_\tau}{s^{n/2}} e^{-s\bar{L}(q)} \leq K(s, 0, -qs) \leq \frac{C_\tau}{s^{n/2}} e^{-s\bar{L}(q)} \quad \text{for all } q \in V_0, s \geq 1.$$

In particular, (18) applies with $s = 1/\varepsilon$ for every $\varepsilon \in (0, 1]$.

Next, the lower quadratic bound in Corollary 2.2, together with the Lipschitz continuity of g and the estimate $|\bar{y}| \leq C$, implies that after enlarging C if necessary, one has

$$h(y) - h(\bar{y}) \geq \frac{1}{4} |y - \bar{y}|^2 \quad \text{whenever } |y - \bar{y}| \geq 10C.$$

Since $|\bar{y}| \leq C$, this yields

$$(19) \quad h(y) - h(\bar{y}) \geq \frac{|y|^2}{8} \quad \text{for } |y| \geq 10C.$$

Using (11), when C is sufficiently large,

$$\int_{\mathbb{R}^n \setminus B(0, 10C)} K\left(\frac{1}{\varepsilon}, 0, \frac{y}{\varepsilon}\right) e^{-\frac{g(y)}{\varepsilon}} dy \leq \frac{1}{4} \int_{B(0, 1)} K\left(\frac{1}{\varepsilon}, 0, \frac{y}{\varepsilon}\right) e^{-\frac{g(y)}{\varepsilon}} dy$$

Accordingly,

$$\int_{\mathbb{R}^n} K\left(\frac{1}{\varepsilon}, 0, \frac{y}{\varepsilon}\right) e^{-\frac{g(y)}{\varepsilon}} dy = r_\varepsilon \int_{B(0, 10C)} K\left(\frac{1}{\varepsilon}, 0, \frac{y}{\varepsilon}\right) e^{-\frac{g(y)}{\varepsilon}} dy$$

for $r_\varepsilon \in [1, \frac{4}{3}]$.

Hence, replacing $\mathbb{R}^n$ by $B(0, 10C)$ changes $u^\varepsilon(0, 1)$ by at most $O(\varepsilon)$, which is absorbed in the final error term.

Owing to (10) and (18), we have

$$\begin{aligned} u^\varepsilon(0, 1) &= -\varepsilon \log \left(\varepsilon^{-n} \int_{\mathbb{R}^n} K\left(\frac{1}{\varepsilon}, 0, \frac{y}{\varepsilon}\right) e^{-\frac{g(y)}{\varepsilon}} dy \right) \\ &= -\varepsilon \log \left(\varepsilon^{-n} \int_{B(0, 10C)} K\left(\frac{1}{\varepsilon}, 0, \frac{y}{\varepsilon}\right) e^{-\frac{g(y)}{\varepsilon}} dy \right) + O(\varepsilon) \\ &= -\varepsilon \log \left(\varepsilon^{-\frac{n}{2}} \int_{B(0, 10C)} e^{-\frac{\bar{L}(-y) + g(y)}{\varepsilon}} dy \right) + O(\varepsilon). \end{aligned}$$

Therefore,

$$(20) \quad u^\varepsilon(0, 1) - u(0, 1) = -\varepsilon \log \left(\varepsilon^{-\frac{n}{2}} \int_{B(0, 10C)} e^{-\frac{h(y) - h(\bar{y})}{\varepsilon}} dy \right) + O(\varepsilon).$$

For $\varepsilon \in (0, 1]$, we use (17) to see that

$$C_1 \geq \int_{B(0, 10C)} e^{-\frac{h(y) - h(\bar{y})}{\varepsilon}} dy \geq \int_{B(\bar{y}, 9C)} e^{-\frac{A|y - \bar{y}|}{\varepsilon}} dy \geq C_2 \varepsilon^n$$

for some positive constants $C_1 = C_1(n)$ and $C_2 = C_2(\|Dg\|_{L^\infty}, \|DV\|_{L^\infty}, n)$.

Hence,

$$|u^\varepsilon(0, 1) - u(0, 1)| \leq \varepsilon \left(\frac{n}{2} |\log \varepsilon| + C \right).$$

□

3.2. Proof of Theorem 1.2. Recall that, by the Hopf–Lax formula,

$$u(x, 1) = \min_{y \in \mathbb{R}^n} \{g(y) + \bar{L}(x - y)\}$$

Therefore, $u(\cdot, 1)$ is semiconcave. For each $x \in \mathbb{R}^n$, denote by $y_x \in \mathbb{R}^n$ such that

$$u(x, 1) = g(y_x) + \bar{L}(x - y_x).$$

If $u(\cdot, 1)$ is differentiable at x , we have that

$$Du(x, 1) = D\bar{L}(x - y_x).$$

Accordingly,

$$(21) \quad y_x = x - D\bar{H}(Du(x, 1)).$$

Write

$$h_x(y) = g(y) + \bar{L}(x - y).$$

The following lemma was supplied to us by ChatGPT. If the initial data g is C^2 , it can be established by adapting the proof of [24, Proposition 4.4(ii)]. See Remark 3.6 for details.

Lemma 3.5. *Suppose that $u(\cdot, 1)$ is twice differentiable at $x = x_0$ and write*

$$y_{x_0} = y_0 \quad \text{and} \quad h_{x_0} = h_0.$$

Owing to (21), y_0 is the unique minimum point of h_0 . Then, there exist $\delta_{x_0}, r_{x_0} > 0$ such that

$$h_0(y) \geq h_0(y_0) + \delta_{x_0} |y - y_0|^2 \quad \text{for } y \in B(y_0, r_{x_0}).$$

Proof. Since this is a local conclusion, without loss of generality, we may assume that

$$I_n \leq D^2 \bar{L} \leq K I_n$$

for a constant $K > 0$.

Set

$$\phi(y) := h_0(y) - h_0(y_0) > 0 \quad \text{for } y \neq y_0.$$

We argue by contradiction. Suppose that the conclusion fails. Then there exists a sequence $y_k \rightarrow y_0$ such that $y_k \neq y_0$ and

$$\phi(y_k) = o(|y_k - y_0|^2).$$

Write

$$d_k := y_k - y_0, \quad \phi_k := \phi(y_k).$$

Since $u(\cdot, 1)$ is twice differentiable at x_0 ,

$$u(x, 1) = u(x_0, 1) + Du(x_0, 1) \cdot (x - x_0) + O(|x - x_0|^2) \quad \text{as } x \rightarrow x_0.$$

Also,

$$Du(x_0, 1) = D\bar{L}(x_0 - y_0).$$

Set

$$p_0 := D\bar{L}(x_0 - y_0), \quad q_k := p_0 - D\bar{L}(x_0 - y_k).$$

Because $I_n \leq D^2\bar{L} \leq KI_n$, we have

$$|d_k| \leq |q_k| \leq K|d_k|.$$

Now define

$$s_k := \frac{2\phi_k}{|d_k|}, \quad x_k := x_0 + s_k \frac{q_k}{|q_k|}.$$

Since $\phi_k = o(|d_k|^2)$, it follows that $s_k = o(|d_k|)$, hence $x_k \rightarrow x_0$.

Using y_k as a competitor in the definition of $u(x_k, 1)$, we obtain

$$u(x_k, 1) \leq g(y_k) + \bar{L}(x_k - y_k).$$

Therefore,

$$u(x_k, 1) - u(x_0, 1) - p_0 \cdot (x_k - x_0) \leq \phi_k + \left(\bar{L}(x_k - y_k) - \bar{L}(x_0 - y_k) - p_0 \cdot (x_k - x_0) \right).$$

By Taylor's theorem and the bound $D^2\bar{L} \leq KI_n$,

$$\bar{L}(x_k - y_k) - \bar{L}(x_0 - y_k) \leq D\bar{L}(x_0 - y_k) \cdot (x_k - x_0) + \frac{K}{2}|x_k - x_0|^2.$$

Hence

$$u(x_k, 1) - u(x_0, 1) - p_0 \cdot (x_k - x_0) \leq \phi_k + (D\bar{L}(x_0 - y_k) - p_0) \cdot (x_k - x_0) + \frac{K}{2}|x_k - x_0|^2.$$

Since $D\bar{L}(x_0 - y_k) - p_0 = -q_k$ and $x_k - x_0 = s_k q_k / |q_k|$, this yields

$$u(x_k, 1) - u(x_0, 1) - p_0 \cdot (x_k - x_0) \leq \phi_k - s_k |q_k| + \frac{K}{2}s_k^2.$$

Using $|q_k| \geq |d_k|$ and $s_k = 2\phi_k / |d_k|$, we get

$$u(x_k, 1) - u(x_0, 1) - p_0 \cdot (x_k - x_0) \leq -\phi_k + \frac{2K\phi_k^2}{|d_k|^2}.$$

Since $\phi_k = o(|d_k|^2)$, it follows that for k large,

$$u(x_k, 1) - u(x_0, 1) - p_0 \cdot (x_k - x_0) \leq -\frac{1}{2}\phi_k.$$

On the other hand, since u is twice differentiable at x_0 ,

$$u(x_k, 1) - u(x_0, 1) - p_0 \cdot (x_k - x_0) = O(|x_k - x_0|^2) = O(s_k^2).$$

But

$$s_k^2 = \frac{4\phi_k^2}{|d_k|^2} = o(\phi_k),$$

which contradicts the previous inequality since $\phi_k > 0$. This proves the claim. $\square$

Remark 3.6. If g is C^2 , the above lemma can be proved by a more straightforward method. The notation “ $\cdot$ ” below represents matrix multiplications. In fact, at $y = y_x$,

$$D^2h(y_x) = D^2g(y_x) + D^2\bar{L}(x - y_x) \geq 0.$$

Also

$$y_x = x - D\bar{H}(Du(x, 1)) \Rightarrow Dy_x = I_n - D^2\bar{H} \cdot D^2u(x, 1).$$

Moreover,

$$Du(x, 1) = Dg(y_x) \Rightarrow D^2u(x, 1) = D^2g \cdot (I_n - D^2\bar{H} \cdot D^2u(x, 1)).$$

Accordingly,

$$(22) \quad (I_n + D^2g \cdot D^2\bar{H}) \cdot D^2u(x, 1) = D^2g(y_x).$$

If $D^2h(y_x)$ is not strictly positive, then there exists a unit vector ξ such that

$$\xi \cdot D^2h(y_x) = \xi \cdot (D^2g(y_x) + D^2\bar{L}(x - y_x)) = 0.$$

Since $D^2\bar{H}(p) \cdot D^2\bar{L}(D\bar{H}(p)) = I_n$, we have

$$\xi \cdot (D^2g(y_x) \cdot D^2\bar{H}(Du(x, 1) + I_n)) = 0.$$

Combining with (22), we derive that

$$\xi \cdot D^2g(y_x) = 0,$$

which implies further that

$$\xi \cdot D^2\bar{L}(x - y_x) = 0,$$

which is impossible due to the strict convexity of $\bar{L}$. ChatGPT helped us find an argument to remove the C^2 assumption.

Proof of Theorem 1.2. Without loss of generality, we assume $(x_0, t_0) = (0, 1)$, and $u(\cdot, 1)$ is twice differentiable at 0.

Since g is locally semiconcave, $h = h_0$ is also locally semiconcave. Therefore, combining with Lemma 3.5, there exists $\alpha_0 = \alpha_{x_0} > 0$ such that

$$\alpha_0|y - \bar{y}|^2 \leq h(y) - h(\bar{y}) \leq \frac{1}{\alpha_0}|y - \bar{y}|^2 \quad \text{for } y \in B(0, 10C).$$

Thanks to (20), we get

$$|u^\varepsilon(0, 1) - u(0, 1)| \leq C_{0,1}\varepsilon.$$

□

Remark 3.7. The local semiconcavity of g is an essential assumption in the proof of Theorem 1.2. Indeed, the following example shows that the statement can fail without it.

Let $V \equiv 0$, $g(x) = \min\{|x|, 10\}$ for $x \in \mathbb{R}^n$. Then, $\bar{H}(p) = \frac{1}{2}|p|^2$ for $p \in \mathbb{R}^n$, and $\bar{L}(q) = \frac{1}{2}|q|^2$ for $q \in \mathbb{R}^n$. By direct computations, we see that

$$u(x, 1) = \frac{|x|^2}{2} \quad \text{for } x \in B(0, 1).$$

In particular, $u(\cdot, 1)$ is smooth at $x_0 = 0$, and $\bar{y} = 0$. Then, for $y \in B(0, 10)$,

$$h(y) - h(0) = g(y) + \bar{L}(-y) = |y| + \frac{|y|^2}{2}.$$

And,

$$\begin{aligned} u^\varepsilon(0, 1) - u(0, 1) &= -\varepsilon \log \left(\varepsilon^{-\frac{n}{2}} \int_{B(0, 10C)} e^{-\frac{h(y)-h(0)}{\varepsilon}} dy \right) + O(\varepsilon) \\ &= \frac{n}{2} \varepsilon |\log \varepsilon| + O(\varepsilon). \end{aligned}$$

Thus, even if $u(\cdot, 1)$ is smooth at $x_0 = 0$, the convergence rate of $u^\varepsilon(0, 1) - u(0, 1)$ is still just $O(\varepsilon |\log \varepsilon|)$.

4. PROOF OF PROPOSITION 3.1

For $p \in \mathbb{R}^n$, let v_p be the unique $\mathbb{Z}^n$ -periodic solution to the cell problem

$$(23) \quad -\frac{1}{2} \Delta v_p + \frac{1}{2} |p + Dv_p|^2 + V(x) = \bar{H}(p) \quad \text{in } \mathbb{R}^n$$

subject to $v_p(0) = 0$.

Throughout this section, $e^{t\mathcal{L}}g$ represents the solution to

$$\begin{cases} u_t = \mathcal{L}u = \frac{1}{2} \Delta u + Vu & \text{in } \mathbb{R}^n \times (0, \infty), \\ u(x, 0) = g(x) & \text{on } \mathbb{R}^n. \end{cases}$$

4.1. Doob h -transform.

Lemma 4.1. *Let $h_p(x) = e^{-p \cdot x - v_p(x)}$ for $x \in \mathbb{R}^n$. Then,*

$$K(t, x, y) = e^{t\bar{H}(p)} \frac{h_p(x)}{h_p(y)} \tilde{p}_p(t, x, y) \quad \text{for } (t, x, y) \in (0, \infty) \times \mathbb{R}^n \times \mathbb{R}^n.$$

Here, $\tilde{p}_p(t, x, y)$ is the fundamental solution to $\partial_t - \mathcal{G}_p$, where

$$(24) \quad \mathcal{G}_p f = \frac{1}{2} \Delta f + b_p(x) \cdot Df.$$

And, $b_p(x) = -p - Dv_p(x)$, which is $\mathbb{Z}^n$ -periodic.

Proof. Denote by $r_p(y) = e^{-v_p(y)}$. Then

$$\frac{1}{2} \Delta r_p - p \cdot D r_p + \left(\frac{1}{2} |p|^2 + V(y) - \bar{H}(p) \right) r_p = 0 \quad \text{in } \mathbb{T}^n,$$

and $r_p(0) = 1$. Also, $h_p(x) = e^{-p \cdot x} r_p(x)$ for $x \in \mathbb{R}^n$.

Let

$$\varphi(x, t) = e^{t\bar{H}(p)} h_p(x) = \exp(t\bar{H}(p) - p \cdot x - v_p(x)).$$

Then

$$\varphi = e^{t\mathcal{L}} h_p.$$

Let $u(x, t) = (e^{t\mathcal{L}}f)(x)$. Let

$$v(x, t) = \frac{u(x, t)}{\varphi(x, t)}.$$

Then,

$$\begin{aligned} v_t &= \frac{u_t}{\varphi} - u \frac{\varphi_t}{\varphi^2}, \\ Dv &= \frac{Du}{\varphi} - u \frac{D\varphi}{\varphi^2}, \\ \Delta v &= \frac{\Delta u}{\varphi} - u \frac{\Delta \varphi}{\varphi^2} - 2 \left(\frac{Du}{\varphi} - u \frac{D\varphi}{\varphi^2} \right) \cdot \frac{D\varphi}{\varphi} \\ &= \frac{\Delta u}{\varphi} - u \frac{\Delta \varphi}{\varphi^2} - 2Dv \cdot (-p - Dv_p). \end{aligned}$$

Hence,

$$\begin{aligned} v_t - \frac{1}{2}\Delta v &= \frac{u_t - \frac{1}{2}\Delta u}{\varphi} - u \frac{\varphi_t - \frac{1}{2}\Delta \varphi}{\varphi^2} + Dv \cdot (-p - Dv_p) \\ &= -\frac{Vu}{\varphi} + u \frac{V\varphi}{\varphi^2} + Dv \cdot (-p - Dv_p) = Dv \cdot (-p - Dv_p). \end{aligned}$$

Thus,

$$v_t - \frac{1}{2}\Delta v = Dv \cdot (-p - Dv_p) = b_p \cdot Dv.$$

Hence,

$$v(x, t) = \int_{\mathbb{R}^n} \tilde{p}_p(t, x, y) \frac{f(y)}{h_p(y)} dy.$$

We conclude that

$$\frac{1}{e^{t\bar{H}(p)}h_p(x)} \int_{\mathbb{R}^n} K(t, x, y) f(y) dy = \int_{\mathbb{R}^n} \tilde{p}_p(t, x, y) \frac{1}{h_p(y)} f(y) dy.$$

Since f is arbitrary,

$$K(t, x, y) = e^{t\bar{H}(p)} \frac{h_p(x)}{h_p(y)} \tilde{p}_p(t, x, y).$$

□

Remark 4.2 (Doob h -transform from the probabilistic point of view). The PDE computation above has a natural probabilistic interpretation. Let $(X_t)_{t \geq 0}$ be Brownian motion in $\mathbb{R}^n$ under $\mathbb{P}_x$, started from x . Since the linear operator here is

$$\mathcal{L} = \frac{1}{2}\Delta + V,$$

the Feynman–Kac formula gives

$$(e^{t\mathcal{L}}f)(x) = \mathbb{E}_x \left[\exp \left(\int_0^t V(X_s) ds \right) f(X_t) \right].$$

Thus, the zeroth-order term $+V$ appears probabilistically as a multiplicative path weight. Depending on the sign convention, such a factor may be viewed as *growth*,

discounting, or killing after a standard renormalization by constants; in any case, the resulting semigroup is no longer conservative, since it does not preserve constants.

Now h_p is a positive eigenfunction satisfying

$$\mathcal{L}h_p = \overline{H}(p) h_p.$$

By Itô's formula, the process

$$M_t^{(p)} := e^{-t\overline{H}(p)} \exp\left(\int_0^t V(X_s) ds\right) \frac{h_p(X_t)}{h_p(x)}$$

is a martingale under $\mathbb{P}_x$. Hence it defines a new probability measure on path space by

$$\frac{d\tilde{\mathbb{P}}_{x,p}}{d\mathbb{P}_x} \Big|_{\mathcal{F}_t} = M_t^{(p)}.$$

Under this new measure, the coordinate process is a genuine diffusion with generator

$$\mathcal{G}_p f = \frac{1}{2} \Delta f + b_p(x) \cdot Df, \quad b_p(x) = D \log h_p(x) = -p - Dv_p(x).$$

In this sense, the Doob h -transform converts the non-conservative Feynman–Kac weighted Brownian motion into a conservative Markov diffusion: the original zeroth-order potential term is absorbed into the change of measure and reappears as the drift b_p .

The kernel identity

$$K(t, x, y) = e^{t\overline{H}(p)} \frac{h_p(x)}{h_p(y)} \tilde{p}_p(t, x, y)$$

is exactly the density-level version of this renormalization. Equivalently, the PDE quotient transformation

$$u = \varphi v, \quad \varphi(x, t) = e^{t\overline{H}(p)} h_p(x),$$

is the analytic counterpart of the same probabilistic mechanism: dividing by the positive solution φ removes the zeroth-order term from the equation and produces a drift-diffusion equation for v ; see also [23].

We have the following large-time bound along the ballistic ray, which is a corollary of [22, Theorem 1.3].

Proposition 4.3 (Local limit bound, two-sided). *Fix a compact set $P \subset \mathbb{R}^n$. There exist $t_0 \geq 1$ and constants $0 < C_1 \leq C_2 < \infty$ such that for every $p \in P$ and every $t \geq t_0$,*

$$(25) \quad C_1 t^{-n/2} \leq \tilde{p}_p(t, 0, -D\overline{H}(p)t) \leq C_2 t^{-n/2}.$$

A sharper local limit asymptotic was announced in [1] and later proved in the more general branching-diffusion setting in [14, Proposition 2.1]. Proposition 4.3 extracts only the uniform two-sided estimate needed for the proof of Proposition 3.1. Appendix A later records a finer asymptotic, formulated for the present family of Doob-transformed periodic diffusions.

Proof. Recall that π_p solves

$$\begin{cases} \mathcal{G}_p^* \pi_p = -\frac{1}{2} \Delta \pi_p + \operatorname{div}(\pi_p b_p) = 0 & \text{in } \mathbb{T}^n, \\ \int_{\mathbb{T}^n} \pi_p(x) dx = 1. \end{cases}$$

To apply the result from [22], we first rewrite the operator $\mathcal{G}_p$ into the so called “canonical form” ([22, equation (1.4)]):

$$\mathcal{G}_p f = \operatorname{div}_{\pi_p} \left(\left(\frac{1}{2} I + \beta(x) \right) Df \right) + \frac{\bar{b}}{\pi_p(x)} \cdot Df.$$

Here,

$$\operatorname{div}_{\pi_p} X = \frac{1}{\pi_p} \operatorname{div}(\pi_p X).$$

denotes the divergence associated with π_p .

Also,

$$\bar{b} = -D\bar{H}(p) = \int_{\mathbb{T}^n} b_p(x) \pi_p(x) dx,$$

and β_p is a periodic antisymmetric matrix field chosen so that

$$J(x) = \pi_p(x) b_p(x) - \frac{1}{2} D\pi_p(x) = \bar{b} + \operatorname{div}(\pi_p \beta_p).$$

More explicitly, $\beta_p = \frac{K(x)}{\pi_p(x)}$ and

$$K_{ij}(x) := (\psi_i)_{x_j} - (\psi_j)_{x_i},$$

where ψ is the unique mean-zero periodic solution to the following Poisson equation

$$-\Delta \psi = -F \quad \text{in } \mathbb{T}^n,$$

where

$$F(x) := J(x) - \int_{\mathbb{T}^n} J(y) dy = J(x) - \bar{b}.$$

Since $\operatorname{div}(J) = 0$, $\operatorname{div}(\psi) = 0$. Moreover, because $p \mapsto v_p$ is continuous in $C^{2,\alpha}(\mathbb{T}^n)$ on compact subsets of $\mathbb{R}^n$, the maps $p \mapsto \pi_p$, $p \mapsto \bar{b}$, and $p \mapsto \beta_p$ are continuous on P . Since P is compact, the canonical-form coefficients above remain in a compact subset of the class covered by [22, Theorem 1.3]. Therefore, the constants in [22, Theorem 1.3] may be chosen uniformly for $p \in P$, and the conclusion follows. $\square$

Remark 4.4. The estimate in Proposition 4.3 is a ballistic local central limit type result, or equivalently a sharp long-time heat-kernel asymptotic, for the periodic diffusion generated by $\mathcal{G}_p$; compare [2, 22]. The Appendix explains the same Gaussian mechanism from a Bloch–Floquet theory/spectral perspective; see also [18, 7, 17].

Proof of Proposition 3.1. Fix a compact set $V_0 \subset \mathbb{R}^n$. For each $q \in V_0$, choose $p \in \mathbb{R}^n$ such that

$$q = D\bar{H}(p).$$

Since $\overline{H}$ is strictly convex and superlinear in the quadratic setting (see, for instance, [26, Chapter 2]), the set

$$P := \{p \in \mathbb{R}^n : D\overline{H}(p) \in V_0\}$$

is compact.

By Lemma 2.1, for every $t > 0$ and $y \in \mathbb{R}^n$,

$$K(t, 0, y) = e^{t\overline{H}(p)} \frac{h_p(0)}{h_p(y)} \tilde{p}_p(t, 0, y),$$

where

$$h_p(x) = e^{-p \cdot x - v_p(x)}.$$

Taking $y = -qt = -D\overline{H}(p)t$, we obtain

$$K(t, 0, -qt) = e^{t\overline{H}(p)} \exp(-p \cdot qt + v_p(0) - v_p(-qt)) \tilde{p}_p(t, 0, -qt).$$

Equivalently,

$$K(t, 0, -qt) = \exp\left(t(\overline{H}(p) - p \cdot q)\right) \exp(v_p(0) - v_p(-qt)) \tilde{p}_p(t, 0, -qt).$$

Now, since $q = D\overline{H}(p)$, by the Legendre duality relation (see [26, Chapter 2]), we have

$$\overline{L}(q) = q \cdot p - \overline{H}(p),$$

and hence

$$\overline{H}(p) - p \cdot q = -\overline{L}(q).$$

Therefore,

$$K(t, 0, -qt) = e^{-t\overline{L}(q)} \exp(v_p(0) - v_p(-qt)) \tilde{p}_p(t, 0, -qt).$$

Because v_p is $\mathbb{Z}^n$ -periodic, v_p is bounded on $\mathbb{R}^n$. Moreover, since $p \in P$ and P is compact, there exists $C \geq 1$ such that

$$C^{-1} \leq \exp(v_p(0) - v_p(-qt)) \leq C$$

for all $p \in P$, $t \geq 1$.

On the other hand, by Proposition 4.3,

$$C_1 t^{-n/2} \leq \tilde{p}_p(t, 0, -qt) \leq C_2 t^{-n/2}$$

for all $p \in P$, $t \geq t_0$.

Combining the last two estimates yields

$$c t^{-n/2} e^{-t\overline{L}(q)} \leq K(t, 0, -qt) \leq C t^{-n/2} e^{-t\overline{L}(q)}$$

for all $q \in V_0$, $t \geq t_0$, where $0 < c \leq C < \infty$ depend only on V_0 .

This is exactly the conclusion of Proposition 3.1. □

APPENDIX A. LARGE-TIME GAUSSIAN ASYMPTOTICS VIA BLOCH–FLOQUET DECOMPOSITION

In this appendix, we give a self-contained proof of the large-time Gaussian asymptotics for the heat kernel in a periodic setting. A closely related asymptotic was announced in [1] and later established in the more general branching-diffusion setting in [14, Proposition 2.1]. This more delicate estimate is not needed for the proof of our main theorems. Nevertheless, for the convenience of the reader, and because both the result and its proof may be useful in other contexts, we provide a direct Bloch–Floquet argument adapted to the operator considered here. Bloch–Floquet decompositions for periodic differential operators are classical; see [18, Chapters 2–3] and [7]. The long-time off-diagonal Gaussian asymptotics are also closely related to perturbative spectral analysis near the principal Bloch branch; compare [17].

A.1. Statement of the theorem. Let

$$A = \frac{1}{2}\Delta + b(x) \cdot D \quad \text{on } \mathbb{R}^n,$$

where $b \in \text{Lip}(\mathbb{R}^n, \mathbb{R}^n)$ is $\mathbb{Z}^n$ -periodic. Let $p(t, x, y)$ denote the heat kernel of A , that is, for

$$\begin{cases} u_t = Au = \frac{1}{2}\Delta u + b(x) \cdot Du & \text{in } \mathbb{R}^n \times (0, \infty), \\ u(x, 0) = g(x) & \text{on } \mathbb{R}^n, \end{cases}$$

we have

$$u(x, t) = \int_{\mathbb{R}^n} p(t, x, y) g(y) dy.$$

Let m be the invariant density on $\mathbb{T}^n = \mathbb{R}^n / \mathbb{Z}^n$, normalized by

$$(26) \quad A^* m = -\frac{1}{2}\Delta m + \text{div}(bm) = 0 \quad \text{in } \mathbb{T}^n, \quad \int_{\mathbb{T}^n} m(x) dx = 1.$$

We keep the notation m for its $\mathbb{Z}^n$ -periodic extension to $\mathbb{R}^n$. Define the effective drift

$$\bar{b} = \int_{\mathbb{T}^n} b(x) m(x) dx.$$

For each $j = 1, \dots, n$, let χ_j be the unique periodic solution of

$$(27) \quad A\chi_j = -(b_j - \bar{b}_j) \quad \text{in } \mathbb{T}^n, \quad \int_{\mathbb{T}^n} \chi_j m dx = 0.$$

The effective diffusion matrix $Q = (Q_{jk})$ is defined by

$$Q_{jk} = \int_{\mathbb{T}^n} (e_j + D\chi_j) \cdot (e_k + D\chi_k) m(x) dx.$$

Theorem A.1 (Large-time Gaussian asymptotics). *The matrix Q is symmetric positive definite. Moreover, there exist constants $C, \eta > 0$ such that for all $t \geq 1$ and all $x, y \in \mathbb{R}^n$,*

$$p(t, x, y) = m(y) \frac{1}{(2\pi t)^{n/2} \sqrt{\det Q}} \exp\left(-\frac{1}{2t}(y - x - \bar{b}t) \cdot Q^{-1}(y - x - \bar{b}t)\right) + R(t, x, y),$$

where

$$|R(t, x, y)| \leq C t^{-(n+1)/2} + C e^{-\eta t}.$$

In particular,

$$p(t, x, y) = m(y) \frac{1}{(2\pi t)^{n/2} \sqrt{\det Q}} \exp\left(-\frac{1}{2t}(y - x - \bar{b}t) \cdot Q^{-1}(y - x - \bar{b}t)\right) + o(t^{-n/2})$$

uniformly in $x, y \in \mathbb{R}^n$ as $t \rightarrow \infty$.

A.2. Proof of Theorem A.1.

Proof. The proof of Theorem A.1 is divided into five steps.

Step 1: Bloch representation of the heat kernel. Let $B = [-\pi, \pi)^n$. For each $\xi \in B$, define the fiber operator

$$(28) \quad A_\xi := e^{-i\xi \cdot x} A e^{i\xi \cdot x} = \frac{1}{2}(D + i\xi)^2 + b(x) \cdot (D + i\xi) \quad \text{on } H^2(\mathbb{T}^n, \mathbb{C}).$$

By the Bloch–Floquet decomposition for periodic differential operators, the Bloch transform $\mathcal{B}$ diagonalizes A as the direct integral

$$\mathcal{B} A \mathcal{B}^{-1} = \int_B^\oplus A_\xi d\xi;$$

see [18, Chapters 2–3] and [7]. Consequently,

$$\mathcal{B} e^{tA} \mathcal{B}^{-1} = \int_B^\oplus e^{tA_\xi} d\xi.$$

For every $t > 0$, the fiber semigroup e^{tA_ξ} admits a jointly continuous kernel $K_\xi(t, \cdot, \cdot)$ on $\mathbb{T}^n \times \mathbb{T}^n$ by standard parabolic regularity on the compact manifold $\mathbb{T}^n$. Therefore

$$(29) \quad p(t, x, y) = \frac{1}{(2\pi)^n} \int_B e^{i\xi \cdot (x-y)} K_\xi(t, \bar{x}, \bar{y}) d\xi,$$

where $\bar{x}, \bar{y}$ denote the classes of x, y in $\mathbb{T}^n$.

Step 2: Principal spectral branch near $\xi = 0$. We next analyze the fiber family A_ξ near the untwisted operator $A_0 = A$.

(i) *The eigenvalue 0 of A is algebraically simple.* Since $A1 = 0$, the constant function 1 is a right eigenfunction of A . Since m solves (26), we also have

$$\int_{\mathbb{T}^n} A f m dx = 0 \quad \text{for every } f \in C^2(\mathbb{T}^n),$$

so m is a left eigenfunction corresponding to the eigenvalue 0.

We claim that $\ker A = \text{span}\{1\}$. Indeed, if $Af = 0$ and $f \in C^2(\mathbb{T}^n)$, then the strong maximum principle for uniformly elliptic operators with bounded coefficients implies that f is constant; see, for instance, [23, Chapter 2]. Thus the geometric multiplicity of the eigenvalue 0 equals 1. To rule out generalized eigenvectors, suppose that $Ag = 1$ for some $g \in C^2(\mathbb{T}^n)$. Integrating against m yields

$$1 = \int_{\mathbb{T}^n} 1 \cdot m dx = \int_{\mathbb{T}^n} A g m dx = 0,$$

a contradiction. Hence the algebraic multiplicity of the eigenvalue 0 is also 1.

Since A generates a Markov semigroup on $C(\mathbb{T}^n)$, we have $s(A) = 0$. We also claim that 0 is the only spectral value of A on the imaginary axis. Indeed, if $A\varphi = \mu\varphi$ with $\Re\mu = 0$, then

$$e^{tA}\varphi = e^{t\mu}\varphi \quad (t > 0).$$

Taking $t = 1$ and arguing exactly as in Step 3(i) below with $\xi = 0$, one sees that $|\varphi|$ must be constant on $\mathbb{T}^n$; positivity improvement of e^A then forces φ to be constant, and therefore $\mu = 0$. Since A has compact resolvent, it follows that there exists $\eta_0 > 0$ such that

$$(30) \quad \sigma(A) \setminus \{0\} \subset \{z \in \mathbb{C} : \Re z \leq -4\eta_0\}.$$

Therefore, by analytic perturbation theory for simple isolated eigenvalues [16, Chapter VII, §1], there exist $r_0 > 0$ and analytic branches

$$\lambda(\xi), \quad \Phi_\xi, \quad \Psi_\xi, \quad |\xi| < r_0,$$

such that

$$A_\xi \Phi_\xi = \lambda(\xi) \Phi_\xi, \quad A_\xi^* \Psi_\xi = \overline{\lambda(\xi)} \Psi_\xi, \quad \lambda(0) = 0,$$

and

$$\Phi_0 \equiv 1, \quad \Psi_0 = m.$$

We choose these branches so that

$$(31) \quad \int_{\mathbb{T}^n} \Phi_\xi(x) m(x) dx = 1 \quad (|\xi| < r_0),$$

and

$$(32) \quad \int_{\mathbb{T}^n} \Psi_\xi(x) \Phi_\xi(x) dx = 1 \quad (|\xi| < r_0).$$

Because the principal part is constant and the lower-order coefficients are Lipschitz, elliptic regularity upgrades the analytic branches above to $C^{2,\alpha}(\mathbb{T}^n)$ for every $\alpha \in (0, 1)$. In particular, all $O(|\xi|)$ and $O(|\xi|^2)$ statements below are uniform on $\mathbb{T}^n$.

(ii) *The first-order term.* Write

$$A_\xi = A + \sum_{j=1}^n \xi_j B_j + \frac{1}{2} \sum_{j,k=1}^n \xi_j \xi_k C_{jk}, \quad B_j = i(D_j + b_j), \quad C_{jk} = -\delta_{jk} I.$$

Differentiating the eigenvalue equation $A_\xi \Phi_\xi = \lambda(\xi) \Phi_\xi$ at $\xi = 0$ in the j -th direction gives

$$A(\partial_{\xi_j} \Phi_\xi|_{\xi=0}) + B_j 1 = (\partial_{\xi_j} \lambda)(0).$$

Pairing with m and using $\int A f m dx = 0$, we obtain

$$(\partial_{\xi_j} \lambda)(0) = \int_{\mathbb{T}^n} B_j 1 m dx = i \int_{\mathbb{T}^n} b_j m dx = i \bar{b}_j.$$

Set

$$\nu_j := \partial_{\xi_j} \Phi_\xi|_{\xi=0}, \quad \chi_j := -i \nu_j.$$

Since (31) implies $\int_{\mathbb{T}^n} \nu_j m dx = 0$, the differentiated eigenvalue equation becomes

$$A\chi_j = -(b_j - \bar{b}_j), \quad \int_{\mathbb{T}^n} \chi_j m dx = 0,$$

which is precisely (27). Therefore

$$(33) \quad \Phi_\xi = 1 + i \sum_{j=1}^n \xi_j \chi_j + O(|\xi|^2), \quad \Psi_\xi = m + O(|\xi|),$$

uniformly on $\mathbb{T}^n$.

(iii) *The quadratic term and the effective diffusion matrix.* Introduce the stationary current

$$J := bm - \frac{1}{2} Dm.$$

Equation (26) is equivalent to

$$\operatorname{div} J = 0 \quad \text{in } \mathbb{T}^n.$$

For $f, g \in C^2(\mathbb{T}^n)$, an integration by parts gives

$$(34) \quad \int_{\mathbb{T}^n} f Ag m dx = -\frac{1}{2} \int_{\mathbb{T}^n} Df \cdot Dg m dx + \int_{\mathbb{T}^n} f J \cdot Dg dx.$$

Differentiating the eigenvalue equation twice at $\xi = 0$ gives

$$A(\partial_{\xi_j \xi_k}^2 \Phi_\xi|_{\xi=0}) + B_j \nu_k + B_k \nu_j + C_{jk} 1 = (\partial_{\xi_j \xi_k}^2 \lambda)(0) + (\partial_{\xi_j} \lambda)(0) \nu_k + (\partial_{\xi_k} \lambda)(0) \nu_j.$$

Pairing with m , using $\int Af m dx = 0$, the identities $(\partial_{\xi_j} \lambda)(0) = i\bar{b}_j$, $\nu_j = i\chi_j$, and the normalization $\int_{\mathbb{T}^n} \nu_j m dx = 0$, we obtain

$$(35) \quad (\partial_{\xi_j \xi_k}^2 \lambda)(0) = -\delta_{jk} - \int_{\mathbb{T}^n} \left(D_j \chi_k + D_k \chi_j + (b_j - \bar{b}_j) \chi_k + (b_k - \bar{b}_k) \chi_j \right) m dx.$$

To simplify the last two terms, use $A\chi_j = -(b_j - \bar{b}_j)$ together with (34):

$$\int_{\mathbb{T}^n} (b_j - \bar{b}_j) \chi_k m dx = - \int_{\mathbb{T}^n} \chi_k A\chi_j m dx = \frac{1}{2} \int_{\mathbb{T}^n} D\chi_k \cdot D\chi_j m dx - \int_{\mathbb{T}^n} \chi_k J \cdot D\chi_j dx.$$

Adding the same identity with j and k interchanged and using $\operatorname{div} J = 0$, we get

$$\int_{\mathbb{T}^n} \left((b_j - \bar{b}_j) \chi_k + (b_k - \bar{b}_k) \chi_j \right) m dx = \int_{\mathbb{T}^n} D\chi_j \cdot D\chi_k m dx,$$

because

$$\int_{\mathbb{T}^n} \chi_k J \cdot D\chi_j dx + \int_{\mathbb{T}^n} \chi_j J \cdot D\chi_k dx = \int_{\mathbb{T}^n} J \cdot D(\chi_j \chi_k) dx = - \int_{\mathbb{T}^n} \operatorname{div} J \chi_j \chi_k dx = 0.$$

Substituting this identity into (35), we find

$$(\partial_{\xi_j \xi_k}^2 \lambda)(0) = - \int_{\mathbb{T}^n} (e_j + D\chi_j) \cdot (e_k + D\chi_k) m dx = -Q_{jk}.$$

Hence

$$(36) \quad \lambda(\xi) = i\bar{b} \cdot \xi - \frac{1}{2} \xi \cdot Q\xi + O(|\xi|^3) \quad (\xi \rightarrow 0).$$

(iv) *Positivity of Q and the sign of $\Re\lambda(\xi)$.* The matrix Q is symmetric by definition. For $\nu \in \mathbb{R}^n$, let $\chi_\nu := \sum_{j=1}^n \nu_j \chi_j$. Then

$$\nu \cdot Q\nu = \int_{\mathbb{T}^n} |\nu + D\chi_\nu|^2 m \, dx.$$

Thus Q is nonnegative definite. If $\nu \cdot Q\nu = 0$, then $D\chi_\nu \equiv -\nu$ on $\mathbb{T}^n$, which is impossible unless $\nu = 0$, because χ_ν is periodic. Therefore Q is positive definite.

Since the linear term in (36) is purely imaginary, there exist $c_0 > 0$ and $r \in (0, r_0)$ such that

$$(37) \quad \Re\lambda(\xi) \leq -c_0|\xi|^2 \quad \text{for } |\xi| < r.$$

Step 3: Spectral gap and semigroup decomposition. We now show that the principal Bloch mode controls the large-time behavior of the fiber semigroup.

(i) *Spectral gap away from $\xi = 0$.* Fix $\tau = 1$ and write $T_\xi := e^{A_\xi}$ on $C(\mathbb{T}^n, \mathbb{C})$. Let $X_t^{\tilde{x}}$ solve

$$dX_t = b(X_t) \, dt + dW_t, \quad X_0^{\tilde{x}} = \tilde{x},$$

for a lift $\tilde{x} \in \mathbb{R}^n$ of $x \in \mathbb{T}^n$. A standard Itô computation shows that

$$(38) \quad (T_\xi f)(x) = \mathbb{E} \left[e^{i\xi \cdot (X_1^{\tilde{x}} - \tilde{x})} f(X_1^{\tilde{x}} \bmod \mathbb{Z}^n) \right].$$

Consequently,

$$(39) \quad |T_\xi f| \leq T_0 |f|.$$

The operator T_0 is compact and positivity improving on $C(\mathbb{T}^n)$, because the torus diffusion has a strictly positive continuous transition density at time 1.

We claim that

$$(40) \quad s(A_\xi) < 0 \quad \text{for every } \xi \in B \setminus \{0\},$$

where $s(A_\xi) = \sup\{\Re z : z \in \sigma(A_\xi)\}$. Since A_ξ has compact resolvent, the spectral bound is attained by an eigenvalue. Suppose, to the contrary, that $s(A_\xi) = 0$ for some $\xi \neq 0$. Then there exist $\mu \in \mathbb{C}$ and $\varphi \in C(\mathbb{T}^n) \setminus \{0\}$ such that

$$A_\xi \varphi = \mu \varphi, \quad \Re \mu = 0.$$

Exponentiating the eigenvalue equation gives

$$T_\xi \varphi = e^\mu \varphi, \quad |e^\mu| = 1.$$

Choose $x_0 \in \mathbb{T}^n$ with $|\varphi(x_0)| = \|\varphi\|_\infty$. Using (39),

$$\|\varphi\|_\infty = |T_\xi \varphi(x_0)| \leq T_0 |\varphi|(x_0) \leq \|\varphi\|_\infty.$$

Hence equality holds throughout. Since T_0 is positivity improving, the equality $T_0 |\varphi|(x_0) = \|\varphi\|_\infty$ forces $|\varphi| \equiv \|\varphi\|_\infty$ on $\mathbb{T}^n$. Equality in the triangle inequality in (38) then implies that the complex-valued random variable

$$e^{i\xi \cdot (X_1^{\tilde{x}_0} - \tilde{x}_0)} \varphi(X_1^{\tilde{x}_0} \bmod \mathbb{Z}^n)$$

has almost surely constant argument. Since the diffusion is uniformly elliptic, $X_1^{\tilde{x}_0}$ has a strictly positive density on $\mathbb{R}^n$. Therefore continuity yields

$$\varphi(y \bmod \mathbb{Z}^n) = C e^{-i\xi \cdot y} \quad \text{for all } y \in \mathbb{R}^n$$

for some $C \neq 0$. Periodicity of φ now gives $e^{-i\xi \cdot k} = 1$ for every $k \in \mathbb{Z}^n$, hence $\xi \in 2\pi\mathbb{Z}^n$. Since $\xi \in [-\pi, \pi]^n$ and $\xi \neq 0$, this is impossible. Thus (40) holds.

By compactness of $B \setminus B_r$ and upper semicontinuity of the spectral bound for the compact-resolvent family $\xi \mapsto A_\xi$, there exists $\eta_1 > 0$ such that

$$(41) \quad s(A_\xi) \leq -2\eta_1 \quad \text{for every } \xi \in B \setminus B_r.$$

(ii) *Spectral decomposition for the near fibers.* By (30), the eigenvalue 0 is separated from the rest of the spectrum of A . After shrinking r if necessary, analytic perturbation theory shows that for every $|\xi| < r$,

$$|\lambda(\xi)| \leq \eta_0 \quad \text{and} \quad \sigma(A_\xi) \setminus \{\lambda(\xi)\} \subset \{z \in \mathbb{C} : \Re z \leq -2\eta_0\}.$$

Let $\Gamma_0 = \{z \in \mathbb{C} : |z| = 2\eta_0\}$. Then the Riesz projection

$$\Pi_\xi = \frac{1}{2\pi i} \int_{\Gamma_0} (z - A_\xi)^{-1} dz$$

has rank one and depends analytically on ξ . By (32),

$$\Pi_\xi f = \Phi_\xi \int_{\mathbb{T}^n} \Psi_\xi(y) f(y) dy,$$

so its kernel equals

$$\Pi_\xi(x, y) = \Phi_\xi(x) \Psi_\xi(y).$$

Let Γ_1 be the boundary of the half-plane strip $\{z \in \mathbb{C} : \Re z \leq -\eta_0\}$, oriented positively around $\sigma(A_\xi) \setminus \{\lambda(\xi)\}$. By the Dunford–Taylor functional calculus,

$$e^{tA_\xi} = e^{t\lambda(\xi)} \Pi_\xi + N_\xi(t), \quad N_\xi(t) = \frac{1}{2\pi i} \int_{\Gamma_1} e^{tz} (z - A_\xi)^{-1} dz.$$

The resolvent is uniformly bounded on Γ_1 for $|\xi| < r$, because Γ_1 stays a positive distance away from the spectrum. Hence

$$(42) \quad e^{tA_\xi} = e^{t\lambda(\xi)} \Pi_\xi + N_\xi(t), \quad \|N_\xi(t)\|_{C(\mathbb{T}^n) \rightarrow C(\mathbb{T}^n)} \leq C e^{-\eta_0 t} \quad (|\xi| < r, t \geq 0).$$

Similarly, (41) yields

$$(43) \quad \|e^{tA_\xi}\|_{C(\mathbb{T}^n) \rightarrow C(\mathbb{T}^n)} \leq C e^{-\eta_1 t} \quad (\xi \in B \setminus B_r, t \geq 0).$$

(iii) *Passage from operator bounds to kernel bounds.* For $t = 1$, the kernels $K_\xi(1, \cdot, \cdot)$ are jointly continuous on the compact set $B \times \mathbb{T}^n \times \mathbb{T}^n$, so

$$M_1 := \sup_{\xi \in B} \sup_{x, y \in \mathbb{T}^n} |K_\xi(1, x, y)| < \infty.$$

For $|\xi| < r$ and $t \geq 1$, define

$$R_\xi(t, x, y) := (N_\xi(t-1)[K_\xi(1, \cdot, y)])(x).$$

Then (42) gives

$$|R_\xi(t, x, y)| \leq C M_1 e^{-\eta_0(t-1)} \leq C e^{-\eta t}$$

for some $\eta \in (0, \min\{\eta_0, \eta_1\})$. Therefore

$$(44) \quad K_\xi(t, x, y) = e^{t\lambda(\xi)} \Phi_\xi(x) \Psi_\xi(y) + R_\xi(t, x, y), \quad |R_\xi(t, x, y)| \leq C e^{-\eta t} \quad (|\xi| < r, t \geq 1).$$

Likewise, if $\xi \in B \setminus B_r$ and $t \geq 1$, then

$$|K_\xi(t, x, y)| = |(e^{(t-1)A_\xi}[K_\xi(1, \cdot, y)])(x)| \leq C e^{-\eta t}.$$

Step 4: Reduction to the principal Gaussian integral. Set

$$z := y - x - \bar{b}t.$$

By (29) and Step 3,

$$p(t, x, y) = I_{\text{near}}(t, x, y) + O(e^{-\eta t}),$$

where

$$I_{\text{near}}(t, x, y) := \frac{1}{(2\pi)^n} \int_{|\xi| < r} e^{i\xi \cdot (x-y)} e^{t\lambda(\xi)} \Phi_\xi(\bar{x}) \Psi_\xi(\bar{y}) d\xi.$$

Using (36), write

$$\lambda(\xi) = i\bar{b} \cdot \xi - \frac{1}{2} \xi \cdot Q\xi + \rho(\xi), \quad |\rho(\xi)| \leq C|\xi|^3 \quad (|\xi| < r).$$

Then

$$(45) \quad I_{\text{near}}(t, x, y) = \frac{1}{(2\pi)^n} \int_{|\xi| < r} e^{-i\xi \cdot z} e^{-\frac{t}{2}\xi \cdot Q\xi} e^{t\rho(\xi)} \Phi_\xi(\bar{x}) \Psi_\xi(\bar{y}) d\xi.$$

We now make three standard replacements.

(a) Replace $\Phi_\xi(\bar{x}) \Psi_\xi(\bar{y})$ by $m(\bar{y})$. By (33),

$$\Phi_\xi(\bar{x}) \Psi_\xi(\bar{y}) - m(\bar{y}) = O(|\xi|) \quad \text{uniformly in } x, y.$$

Moreover,

$$\left| e^{-\frac{t}{2}\xi \cdot Q\xi} e^{t\rho(\xi)} \right| = e^{t\Re\lambda(\xi)} \leq e^{-c_0 t|\xi|^2} \quad (|\xi| < r)$$

by (37). Hence the error caused by replacing $\Phi_\xi(\bar{x}) \Psi_\xi(\bar{y})$ with $m(\bar{y})$ is bounded by

$$C \int_{|\xi| < r} |\xi| e^{-c_0 t|\xi|^2} d\xi \leq C t^{-(n+1)/2}.$$

(b) Replace $e^{t\rho(\xi)}$ by 1. Since $|e^w - 1| \leq |w|e^{|w|}$, we obtain

$$|e^{t\rho(\xi)} - 1| \leq C t |\xi|^3 e^{C t |\xi|^3}.$$

After shrinking r once more if necessary, we have $C t |\xi|^3 \leq \frac{c_0}{2} t |\xi|^2$ for $|\xi| < r$, and therefore

$$|e^{t\rho(\xi)} - 1| \leq C t |\xi|^3 e^{\frac{c_0}{2} t |\xi|^2}.$$

Thus the corresponding error is bounded by

$$C \int_{|\xi| < r} t |\xi|^3 e^{-\frac{c_0}{2} t |\xi|^2} d\xi \leq C t^{-(n+1)/2}.$$

(c) *Extend the integral from $|\xi| < r$ to all of $\mathbb{R}^n$.* Since Q is positive definite, there exists $\mu > 0$ such that

$$\xi \cdot Q\xi \geq \mu|\xi|^2 \quad \text{for all } \xi \in \mathbb{R}^n.$$

Hence

$$\int_{|\xi| \geq r} e^{-\frac{t}{2}\xi \cdot Q\xi} d\xi \leq C e^{-\mu r^2 t/4},$$

which is exponentially small.

Combining (a), (b), and (c), we obtain

$$(46) \quad p(t, x, y) = \frac{m(\bar{y})}{(2\pi)^n} \int_{\mathbb{R}^n} e^{-i\xi \cdot z} e^{-\frac{t}{2}\xi \cdot Q\xi} d\xi + O(t^{-(n+1)/2}) + O(e^{-\eta t})$$

uniformly in $x, y \in \mathbb{R}^n$.

Step 5: Evaluation of the Gaussian Fourier integral. The standard Gaussian Fourier inversion formula gives

$$\frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} e^{-i\xi \cdot z} e^{-\frac{t}{2}\xi \cdot Q\xi} d\xi = \frac{1}{(2\pi t)^{n/2} \sqrt{\det Q}} \exp\left(-\frac{1}{2t} z \cdot Q^{-1} z\right).$$

Substituting this into (46) and using the periodicity $m(\bar{y}) = m(y)$, we obtain the asserted asymptotic for all sufficiently large t .

It remains to extend the estimate to every $t \geq 1$. Fix $t_0 \geq 1$ so that the preceding argument is valid for $t \geq t_0$. For $t \in [1, t_0]$, formula (29) and compactness of $[1, t_0] \times B \times \mathbb{T}^n \times \mathbb{T}^n$ imply that

$$\sup_{1 \leq t \leq t_0} \sup_{x, y \in \mathbb{R}^n} |p(t, x, y)| < \infty.$$

The Gaussian main term is also uniformly bounded on $[1, t_0] \times \mathbb{R}^n \times \mathbb{R}^n$. Enlarging the constant C if necessary therefore yields the desired estimate for all $t \geq 1$. The proof is complete. $\square$

A.3. Two consequences. The proof of Theorem A.1 is stable under compact perturbations of the periodic drift. We record the resulting uniform version, which is the form needed in our application to the Doob-transformed diffusion.

Theorem A.2 (Uniform compact-family version). *Let $\mathcal{P}$ be a compact parameter space, and assume that*

$$\alpha \longmapsto b^{(\alpha)}$$

is continuous from $\mathcal{P}$ into $\text{Lip}(\mathbb{T}^n, \mathbb{R}^n)$. For each $\alpha \in \mathcal{P}$, let

$$A^{(\alpha)} = \frac{1}{2}\Delta + b^{(\alpha)}(x) \cdot D$$

and denote by $p_\alpha(t, x, y)$ its heat kernel on $\mathbb{R}^n$. Let m_α , $\bar{b}_\alpha$, $\chi_{\alpha,j}$, and Q_α be the invariant density, effective drift, correctors, and effective diffusion matrix associated with $A^{(\alpha)}$. Then there exist constants $C_{\mathcal{P}, \eta_{\mathcal{P}}} > 0$ such that for every $\alpha \in \mathcal{P}$, every $t \geq 1$, and every $x, y \in \mathbb{R}^n$,

$$p_\alpha(t, x, y) = m_\alpha(y) \frac{1}{(2\pi t)^{n/2} \sqrt{\det Q_\alpha}} \exp\left(-\frac{1}{2t}(y - x - \bar{b}_\alpha t) \cdot Q_\alpha^{-1}(y - x - \bar{b}_\alpha t)\right) \\ + R_\alpha(t, x, y),$$

with

$$|R_\alpha(t, x, y)| \leq C_{\mathcal{P}} t^{-(n+1)/2} + C_{\mathcal{P}} e^{-\eta_{\mathcal{P}} t}.$$

Proof. We now verify that the five-step proof above can be carried out uniformly on compact parameter sets.

Uniform elliptic bounds and continuous dependence. Since $\mathcal{P}$ is compact and $\alpha \mapsto b^{(\alpha)}$ is continuous in Lip, there exists $M > 0$ such that

$$\sup_{\alpha \in \mathcal{P}} \|b^{(\alpha)}\|_{\text{Lip}(\mathbb{T}^n)} \leq M.$$

Hence, all elliptic and parabolic estimates below are uniform in α . Standard Schauder theory, together with uniqueness and the Fredholm alternative on the torus, implies that

$$\alpha \mapsto m_\alpha, \quad \alpha \mapsto \chi_{\alpha,j}$$

are continuous from $\mathcal{P}$ into $C^{2,\beta}(\mathbb{T}^n)$ for every $\beta \in (0, 1)$. In particular,

$$\alpha \mapsto \bar{b}_\alpha, \quad \alpha \mapsto Q_\alpha$$

are continuous on $\mathcal{P}$, and $\inf_{\alpha \in \mathcal{P}} \det Q_\alpha > 0$.

Uniform Step 1. For each $\alpha \in \mathcal{P}$ and $\xi \in B$, define

$$A_{\alpha,\xi} := e^{-i\xi \cdot x} A^{(\alpha)} e^{i\xi \cdot x}.$$

The map $(\alpha, \xi) \mapsto A_{\alpha,\xi}$ is continuous from $\mathcal{P} \times B$ into $\mathcal{L}(H^2(\mathbb{T}^n), L^2(\mathbb{T}^n))$. By parabolic regularity, the kernels $K_{\alpha,\xi}(t, \cdot, \cdot)$ of $e^{tA_{\alpha,\xi}}$ are jointly continuous, and for each fixed $t > 0$,

$$(\alpha, \xi, x, y) \mapsto K_{\alpha,\xi}(t, x, y)$$

is continuous on the compact set $\mathcal{P} \times B \times \mathbb{T}^n \times \mathbb{T}^n$. In particular,

$$\sup_{\alpha \in \mathcal{P}} \sup_{\xi \in B} \sup_{x, y \in \mathbb{T}^n} |K_{\alpha,\xi}(1, x, y)| < \infty.$$

Uniform Steps 2 and 3. For each $\alpha \in \mathcal{P}$, the proof of Theorem A.1 gives a simple isolated eigenvalue 0 for $A^{(\alpha)}$, an analytic principal branch $\lambda_\alpha(\xi)$, and a positive-definite matrix Q_α . Because $(\alpha, \xi) \mapsto A_{\alpha,\xi}$ is continuous and $\mathcal{P}$ is compact, the contour argument in Step 3 may be chosen uniformly in α : there exist $r \in (0, 1)$, $\eta > 0$, and $C \geq 1$ such that, for every $\alpha \in \mathcal{P}$, we have

1. the branch $\lambda_\alpha(\xi)$ is analytic for $|\xi| < r$, and

$$\lambda_\alpha(\xi) = i\bar{b}_\alpha \cdot \xi - \frac{1}{2} \xi \cdot Q_\alpha \xi + \rho_\alpha(\xi), \quad |\rho_\alpha(\xi)| \leq C|\xi|^3 \quad (|\xi| < r);$$

2. the associated right and left eigenfunctions satisfy

$$\Phi_{\alpha,\xi} = 1 + i \sum_{j=1}^n \xi_j \chi_{\alpha,j} + O(|\xi|^2), \quad \Psi_{\alpha,\xi} = m_\alpha + O(|\xi|),$$

uniformly in $(\alpha, \xi) \in \mathcal{P} \times B_r$;

3. the remainder estimates in Step 3 hold with constants independent of α :

$$K_{\alpha,\xi}(t, x, y) = e^{t\lambda_\alpha(\xi)} \Phi_{\alpha,\xi}(x) \Psi_{\alpha,\xi}(y) + R_{\alpha,\xi}(t, x, y), \quad |R_{\alpha,\xi}(t, x, y)| \leq C e^{-\eta t}$$

for $|\xi| < r$, $t \geq 1$, and

$$|K_{\alpha,\xi}(t, x, y)| \leq C e^{-\eta t}$$

for $\xi \in B \setminus B_r$, $t \geq 1$.

To justify the uniform far- ξ spectral gap, one may argue by contradiction exactly as in the proof of (40): if such a gap failed, there would exist sequences $\alpha_\ell \in \mathcal{P}$, $\xi_\ell \in B \setminus B_r$, and normalized eigenfunctions φ_ℓ of $A_{\alpha_\ell, \xi_\ell}$ with eigenvalues whose real parts converge to 0. Compactness of $\mathcal{P} \times (B \setminus B_r)$ and elliptic compactness would then yield a nontrivial limit eigenfunction for some $A_{\alpha, \xi}$ with $\xi \neq 0$ and spectral bound 0, contradicting the single-parameter argument above. The uniform isolation of the principal eigenvalue near $\xi = 0$ is obtained in the same way.

Uniform Steps 4 and 5. With the uniform constants established above, the three replacements in Step 4 and the Gaussian inversion in Step 5 are unchanged. Every estimate is uniform in α , because the constants controlling Q_α , $\det Q_\alpha$, the remainder ρ_α , and the fiber-kernel decomposition are uniform on $\mathcal{P}$. Therefore the conclusion of Theorem A.1 holds with constants independent of α . $\square$

We now return to the original Schrödinger kernel appearing in Section 4. Combining the Doob h -transform with Theorem A.2 yields a refined ballistic asymptotic with an explicit periodic amplitude.

Proposition A.3 (Sharp ballistic asymptotics for the original Schrödinger kernel). *Keep the notation of Section 4. Let $V_0 \subset \mathbb{R}^n$ and $R_0 \subset \mathbb{R}^n$ be compact. Then there exist constants $C, \eta > 0$ such that, for every $q \in V_0$, every $r \in R_0$, and every $t \geq 1$,*

$$\begin{aligned} & K(t, 0, -qt + r) \\ &= e^{-t\bar{L}(q)} \exp(p(q) \cdot r + v_{p(q)}(-qt + r) - v_{p(q)}(0)) \\ &\times \left[\pi_{p(q)}(-qt + r) \frac{1}{(2\pi t)^{n/2} \sqrt{\det Q_{p(q)}}} \exp\left(-\frac{1}{2t} r \cdot Q_{p(q)}^{-1} r\right) + O(t^{-(n+1)/2}) + O(e^{-\eta t}) \right], \end{aligned}$$

where $p(q)$ is the unique vector satisfying $D\bar{H}(p(q)) = q$, and $Q_{p(q)}$ is the effective diffusion matrix of the periodic diffusion generated by

$$\mathcal{G}_{p(q)} = \frac{1}{2} \Delta + (-p(q) - Dv_{p(q)}(x)) \cdot D.$$

Moreover, in the present mechanical setting, one has $Q_p = D^2\bar{H}(p) = (D^2\bar{L}(q))^{-1}$; this is the standard identification between the effective diffusion matrix of the Doob-transformed process and the Hessian of the effective Hamiltonian, together with the Legendre duality.

In particular, for $r = 0$,

$$K(t, 0, -qt) = e^{-t\bar{L}(q)} a_{p(q)}(-qt) t^{-n/2} + O\left(e^{-t\bar{L}(q)} t^{-(n+1)/2}\right),$$

where the periodic amplitude is

$$a_p(y) := e^{v_p(y) - v_p(0)} \pi_p(y) \frac{1}{(2\pi)^{n/2} \sqrt{\det Q_p}}.$$

The error term is uniform for $q \in V_0$.

Remark A.4. Proposition A.3 is not a direct restatement of [14, Theorem 2.2]. The latter gives a powerful large-deviation asymptotic for the branching-diffusion kernel $u(t, x, y)$ in periodic media, expressed in terms of the rate function Φ and the associated tilted principal eigenfunctions. Our formulation is tailored to the present homogenization problem: it is stated directly for the original Schrödinger kernel K along the ballistic rays $y = -qt + r$, with the prefactor written in the variables naturally associated with the cell problem, namely $p(q)$, $v_{p(q)}$, $\pi_{p(q)}$, and $Q_{p(q)}$, uniformly for $q \in V_0$ and $r \in R_0$. We also note that our proof is organized somewhat differently, through the uniform compact-family Gaussian asymptotics of Theorem A.2 together with the Doob h -transform, so as to fit the proof of Theorem 1.1.

Proof. By the Doob h -transform identity proved in Section 4,

$$K(t, 0, y) = e^{t\bar{H}(p)} \frac{h_p(0)}{h_p(y)} \tilde{p}_p(t, 0, y), \quad h_p(x) = e^{-p \cdot x - v_p(x)}.$$

Taking $y = -qt + r$ and using $q = D\bar{H}(p)$, we obtain

$$K(t, 0, -qt + r) = e^{-t\bar{L}(q)} e^{p \cdot r + v_p(-qt + r) - v_p(0)} \tilde{p}_p(t, 0, -qt + r).$$

Now let

$$P := \{p \in \mathbb{R}^n : D\bar{H}(p) \in V_0\}.$$

Since $\bar{H}$ is strictly convex and superlinear in the quadratic setting, $D\bar{H} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a homeomorphism onto $\mathbb{R}^n$; see [26, Chapter 2]. Hence P is compact, and $q \mapsto p(q)$ is continuous on V_0 .

For each $p \in P$, consider the periodic diffusion with generator

$$\mathcal{G}_p = \frac{1}{2} \Delta + (-p - Dv_p(x)) \cdot D.$$

Standard elliptic regularity for the cell problem (23) implies that $p \mapsto v_p$ is continuous from P into $C^{2,\beta}(\mathbb{T}^n)$ for every $\beta \in (0, 1)$. Therefore

$$p \mapsto b^{(p)}(x) := -p - Dv_p(x)$$

is continuous from P into $\text{Lip}(\mathbb{T}^n, \mathbb{R}^n)$. Theorem A.2 thus applies to the family $\{\mathcal{G}_p\}_{p \in P}$.

For this family, the effective drift equals

$$\bar{b}_p = \int_{\mathbb{T}^n} (-p - Dv_p) \pi_p dx = -D\bar{H}(p) = -q.$$

Thus, with $x = 0$ and $y = -qt + r$,

$$y - x - \bar{b}_p t = r.$$

Applying Theorem A.2 gives

$$\begin{aligned} & \tilde{p}_p(t, 0, -qt + r) \\ &= \pi_p(-qt + r) \frac{1}{(2\pi t)^{n/2} \sqrt{\det Q_p}} \exp\left(-\frac{1}{2t} r \cdot Q_p^{-1} r\right) + O(t^{-(n+1)/2}) + O(e^{-\eta t}), \end{aligned}$$

uniformly for $p \in P$, $r \in R_0$, and $t \geq 1$. Substituting this into the Doob-transform identity proves the first assertion. The case $r = 0$ follows immediately, since the exponentially small term may be absorbed into $O(t^{-(n+1)/2})$ for $t \geq 1$. $\square$

Remark A.5. We recall that $\pi_p = C_p r_p r_{-p}$, where the constant $C_p > 0$ is chosen so that $\int_{\mathbb{T}^n} \pi_p dx = 1$. Hence

$$\begin{aligned} & e^{-t\bar{L}(q)} \exp(p \cdot r + v_p(-qt + r) - v_p(0)) \pi_p(-qt + r) \exp\left(-\frac{1}{2t} r \cdot Q_p^{-1} r\right) \\ &= C_p \exp\left(-t \left(\bar{L}(q) - D\bar{L}(q) \cdot \frac{r}{t} + \frac{1}{2} \frac{r}{t} \cdot D^2 \bar{L}(q) \frac{r}{t} \right)\right) \exp(-v_p(0) - v_{-p}(-qt + r)). \end{aligned}$$

Since $\bar{L} \in C^2(\mathbb{R}^n)$, Taylor's theorem gives

$$\bar{L}\left(q - \frac{r}{t}\right) = \bar{L}(q) - D\bar{L}(q) \cdot \frac{r}{t} + \frac{1}{2} \frac{r}{t} \cdot D^2 \bar{L}(q) \frac{r}{t} + O(t^{-3}),$$

uniformly for $q \in V_0$ and $r \in R_0$. Therefore Proposition A.3 also yields

$$\begin{aligned} & K(t, 0, -qt + r) = \\ & C_p \exp\left(-t\bar{L}\left(q - \frac{r}{t}\right)\right) \exp(-v_p(0) - v_{-p}(-qt + r)) \left[\frac{\sqrt{\det D^2 \bar{L}(q)}}{(2\pi t)^{n/2}} + O(t^{-(n+1)/2}) \right], \end{aligned}$$

where the extra $O(t^{-2})$ produced by Taylor's theorem has been absorbed into the displayed remainder.

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(Z. Liu) SHANGHAI INSTITUTE FOR MATHEMATICS AND INTERDISCIPLINARY SCIENCES (SIMIS), SHANGHAI, CHINA, 200433

(Z. Liu) RESEARCH INSTITUTE OF INTELLIGENT COMPLEX SYSTEMS, FUDAN UNIVERSITY, SHANGHAI 200433, CHINA

Email address: `zliu@simis.cn`

(H. V. Tran) DEPARTMENT OF MATHEMATICS, UNIVERSITY OF WISCONSIN-MADISON, VAN VLECK HALL, 480 LINCOLN DRIVE, MADISON, WISCONSIN 53706, USA

Email address: `hung@math.wisc.edu`

(Y. Yu) DEPARTMENT OF MATHEMATICS, UNIVERSITY OF CALIFORNIA AT IRVINE, CALIFORNIA 92697, USA

Email address: `yifengy@uci.edu`